## Concept 1 - Objects of the IT Act

a) To provide legal recognition to electronic transactions by recognizing digital signatures, whether used by public or by government officials or agencies, including the publication of rules or any other matters, such as gazette notifications.
b) Facilitate electronic filing of documents and retention thereof in govt. records
c) Subsequent Amendments to Other Acts
d) to set up licensing & monitoring of Certifying Authority.

## Concept 2 – Dispatch and Receipt of Records

Dispatch – Records sent by originator, his agent, system programmed by the originator which operates automatically. Dispatch would occur when it enters a resource outside the control of the originator

Receipt – unless otherwise agreed by the parties, receipt would occur when it enters the designated computer resource or else when it is retrieved by the addressee

Place of Dispatch & Place of Receipt - unless otherwise agreed by the parties place of dispatch shall be deemed to be place of business, same in case of receipt.

## Concept 3 - Controller of Certifying Authority (CCA)

The Controller of Certifying Authority (CCA) is to be appointed by Central Government to recognize, license, regulate, standardize, supervise the certifying authorities. Certifying authorities, who are licensed by CCA shall issue digital signature certificate and perform other acts as specified by the Controller and perform other functions as decided by Controller.

## Concept 4 - Settlement of disputes

a) Central Govt. has appointed adjudicating authorities having legal knowledge.
b) Cyber Regulations Appellate Tribunal has also been constituted.
c) Any person not satisfied with the order of controller or adjudicating authorities may appeal within 45 days.
d) Decision of appellate tribunal can be appealed to High Court within 60 days

## Concept 5 – Introduction to Data Protection Laws

Data Protection Laws refers to the set of privacy laws, policies and procedures that aim to minimise intrusion into one's privacy caused by the collection, storage, etc. of personal data. Personal data generally refers to the information or data which relate to a person who can be identified from that information or data whether collected by any Government or any private organization or an agency.

## Concept 6 - Data Protection Law in India

India currently is not having a separate data protection law.

Section 43A & Section 72A were inserted in the IT Act through the Information Technology (Amendment) Act, 2008 and the Central Government, notified the Information Technology (Reasonable security practices and procedures and sensitive personal data or information) Rules, 2011. The right to privacy in India was declared a fundamental right by the Hon'ble Supreme Court of India on August 24, 2017.

Section 43A of the IT Act explicitly provides that whenever a corporate body possesses or deals with any sensitive personal data or information, and is negligent in maintaining a reasonable security to protect such data or information, which thereby causes wrongful loss or wrongful gain to any person, then such body corporate shall be liable to pay damages to the person so affected.

Section 72A ~~provides for the punishment for disclosure of information in breach of lawful contract and any person may be punished with imprisonment for a term not exceeding three years (or) with a fine not exceeding five lakh rupees (or) with both in case disclosure of information is made in breach of lawful contract.~~

Amended on 11th August 2023 - "Newly amended Section 72A" provides for the penalty for disclosure of information in breach of lawful contract and any person may be liable to penalty which may extend to ₹ 25 lakhs in case disclosure of information is made in breach of lawful contract.

## Concept 7 – Main Highlights – Information Technology (Reasonable Security Practices and Procedures and Sensitive Personal Data or Information) Rules, 2011

a) Applicability - Applies to bodies corporate and individuals in India.

b) Sensitive Personal Data (Rule 3)
- Includes passwords, credit/debit card information, biometric data, DNA, fingerprints, mental health condition etc.
- any information is freely available or accessible in the public domain is not considered to be sensitive personal data.

c) Privacy Policy (Rule 4)
- Body corporates that collect sensitive personal data must have a privacy policy.
- Policy must be easily accessible to everyone by publishing on its website.
- The policy must contain details regarding, what kind of personal information is being collected & why, and what security practices are taken to keep it safe and confidential.

d) Data Collection Guidelines (Rule 5)

1) Obtain consent from data providers.
2) Collect data only for lawful and necessary purposes.
3) Use data only for stated purposes and retain it only as long as necessary.
4) Inform data providers about the collection, purpose, and recipients.
5) Allow data providers to review and correct their data.
6) Ensure data security.
7) Designate a Grievance Officer with contact details on the website.

e) Security Practices (Rule 8)
- Implement reasonable security practices, such as IS/ISO/IEC 27001 standards.
- Conduct annual audits or audits upon significant upgrades to security processes

## Concept 8 - Cybercrime - Meaning

Cybercrime involves criminal activities where computers or networks are used as tools, targets, or places of criminal activity, including crimes like credit card fraud, cyber terrorism, etc. Its unique nature, without geographical boundaries and often with anonymous perpetrators, poses significant threats to governments, businesses, and individuals.

## Concept 9 - Cyber Crimes Against Persons

a) Cyber Stalking - Using technology like the internet, email, or phones to threaten and create fear.
b) Obscenity - Sharing indecent or prohibited material, especially child pornography.
c) Defamation - Lowering someone's dignity by hacking their account and sending vulgar emails.
d) Hacking - Unauthorized access to a computer system, often destroying data and programs.
e) Cracking - Breaking into a computer system without permission and tampering with data.
f) Spoofing - Sending emails that misrepresent their origin, often used for spam and identity theft.
g) Carding - Using fake debit and credit cards to steal money from bank accounts.
h) Fraud - Stealing passwords and data with the intent to cheat or deceive.
i) Threat - Using technology to threaten someone's life or safety.

## Concept 10 - Cyber Crimes Against Property

a) IPR Violation - Infringing on intellectual property rights like software piracy and copyright violations.
b) Squatting - Competing for the same domain name, often by registering it first or using a similar one.
c) Vandalism - Deliberately damaging or destroying data and network services.
d) Hacking - Unauthorized access to computer systems, leading to data loss.
e) Virus - Programs that spread and damage data on computers and networks.
f) Trespass - Accessing someone's computer without permission, but not altering or damaging data.

## Concept 11 - Cyber Crimes Against Government

a) Terrorism - Using the internet to conduct attacks that threaten national security.
b) Warfare - Politically motivated hacking to damage or spy, similar to traditional warfare.
c) Piracy - Distributing pirated software to destroy government data and records.
d) unauthorized Information - It is very easy to access any information by the terrorists with the aid of internet and to possess that information for political, religious, social, ideological objectives.

## Concept 12 - Cyber Laws to Protect Against Cyber Crime

Cybercrimes are on the rise due to the extensive use of the internet. The Information Technology (IT) Act, 2000, and amendments to the Indian Penal Code (IPC) aim to combat these crimes. Key offenses made punishable under these laws include:

Cyber Crimes Under the IT Act:

1) Tampering with Computer source documents

2) Hacking with Computer systems, Data alteration
3) Publishing obscene information
4) Unauthorized access to protected system
5) Breach of Confidentiality and Privacy
6) Publishing false digital signature certificates

Cyber Crimes Under IPC and Special Laws:

1) Sending threatening messages by email
2) Defamatory messages by email
3) Forgery of electronic records
4) Bogus websites, cyber frauds
5) Email spoofing
6) Web-Jacking
7) Email Abuse

Cyber Crimes Under Special Acts:

1) Drugs – "Online sale of Drugs under Narcotic Drugs and Psychotropic Substances Act, 1985"
2) Sale of Arms - "Online sale of Arms under Arms Act, 1959"

## Concept 13 – Meaning of Business Intelligence (BI)

Business Intelligence (BI) is a technology-driven process that helps organizations analyse data and provide actionable insights for decision-making. It involves collecting data from internal and external sources, preparing it for analysis, and creating visualizations, dashboards, and reports.

The ultimate goal of BI is to drive better business decisions that enable organizations to increase revenue, improve operational efficiency and gain competitive advantage. To achieve that goal, BI uses a combination of analytics, data management and reporting tools, plus various methodologies.

## Concept 14 – Process of Business Intelligence

a) Data Collection and Storage: BI data is stored in data warehouses or smaller data marts. This data can include both historical information and real-time data from various source systems.
b) Data Preparation: Raw data from different sources is integrated, consolidated, and cleansed using data integration and quality management tools. This ensures accurate and consistent information for analysis.
c) Data Analysis: The prepared data is analysed using analytical querying, where BI tools run queries to extract meaningful insights.
d) Distribution: Key performance indicators (KPIs) and other analytical findings are distributed to business users through reports, dashboards, and visualizations.
e) Decision-Making: The insights and information gathered from the analysis are used to influence and drive strategic and tactical business decisions.

Initially, BI tools were used mainly by BI and IT professionals. However, with the advent of self-service BI tools, business users, including analysts and executives, can now query data, create visualizations, and design dashboards on their own.

Practical Example

"A coffee shop chain uses BI to boost sales. They collect data from their sales systems and customer feedback. The BI tools show which drinks are most popular and which times of the day are busiest. The shop then adjusts its menu and staffing based on these insights. This leads to higher sales and better customer service."

## Concept 15 - Importance of Business Intelligence (BI)

Business Intelligence (BI) is essential for improving business operations by turning data into valuable insights. Organizations using BI tools can make informed, data-driven decisions, leading to increased productivity, revenue, and growth. Without BI, companies rely on intuition and past experiences, which can result in errors and missed opportunities. BI ensures decisions are backed by data, reducing risks and enhancing overall business performance

## Concept 16 - Benefits of Business Intelligence (BI)

a) Speed up and improve decision-making
b) Optimize internal business processes
c) Increase operational efficiency and productivity
d) Spot business problems that need to be addressed
e) Identify emerging business and market trends
f) Develop stronger business strategies
g) Drive higher sales and new revenues
h) Gain a competitive edge over rival companies

## Concept 17 - Types of Business Intelligence (BI) Tools and Applications

1) Ad Hoc Analysis – the process of writing and running queries to analyze specific business issues on temporary basis.
2) Online Analytical Processing (OLAP) – Analysing data in multiple dimensions to handle complex questions and calculations.
3) Mobile BI - Using BI apps on smartphones and tablets, showing simple charts and key metrics.
4) Real-time BI - data is analyzed as it's created, collected and processed to give users an up-to-date view of business operations, customer behaviour, financial markets and other areas of interest
5) Operational Intelligence (OI) - Also called operational BI, this is a form of real-time analytics that delivers information to managers and frontline workers in business operations.
6) Open Source BI (OSBI) Business intelligence software that is open source typically includes two versions: a community edition that can be used free of charge and a subscription-based commercial release with technical support by the vendor.
7) Embedded BI - These tools put BI and data visualization functionality directly into business applications. That enables business users to analyze data within the applications they use to do their job.
8) Collaborative BI - This is more of a process than a specific technology which involves the combination of BI applications and collaboration tools to enable different users to work together on data analysis and share information with one another.

9) Location Intelligence (LI) - This is a specialized form of BI that enables users to analyze location and Geographic data, with map-based data visualization incorporated.

## Concept 18 - Business Intelligence (BI) Platforms

Modern BI platforms typically include:

a) Data Visualization Software - Designs charts and infographics for easy data understanding.
b) BI Dashboards and Reports - Creates dashboards, reports, and scorecards to display key performance indicators (KPIs) and business metrics.
c) Data Storytelling Features - Combines visualizations and text for business presentations.
d) Management Tools - Monitors usage, optimizes performance, and ensures security in BI deployments.

General Use Cases of BI in an enterprise:

a) monitoring business performance
b) supporting decision-making and strategic planning
c) evaluating and improving business processes.
d) giving operational workers useful information about customers, equipment, supply chains etc.
e) detecting trends, patterns and relationships in data.